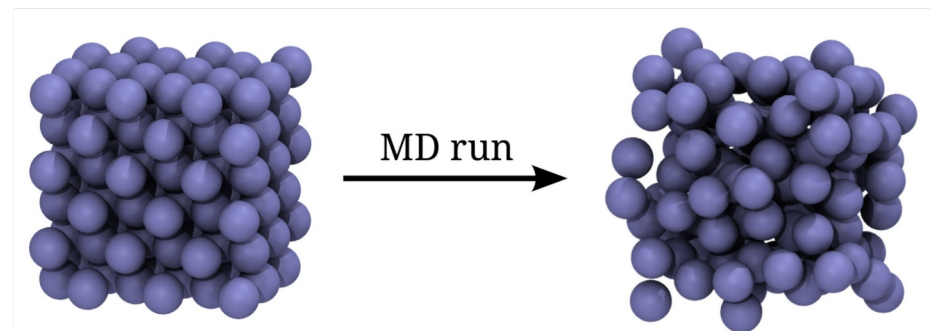


Aula 01 – Introdução ao Python



Prof. Elvis Soares
elvis@peq.coppe.ufrj.br

Infos

Horário de aulas: 3a, 09:00-12:00

Sala de Aula: I224

Calendário: 10/Mar – 07/Jul (~17 encontros)

Professores: Frederico Tavares (tavares@eq.ufrj.br)

Elvis Soares (elvis@peq.coppe.ufrj.br)



<https://github.com/elvissoares/EQE595-SimMol>

CRITÉRIOS DE AVALIAÇÃO

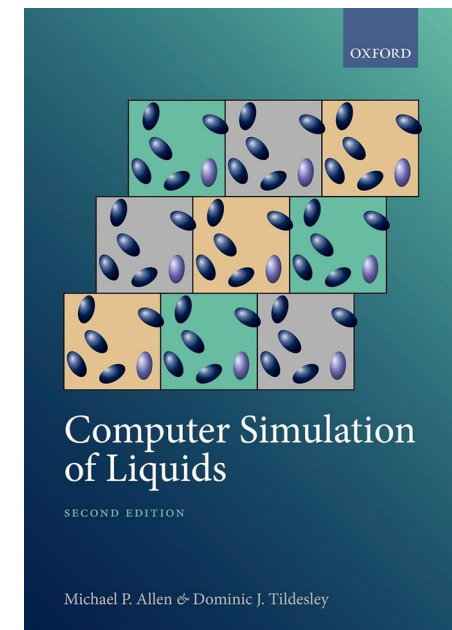
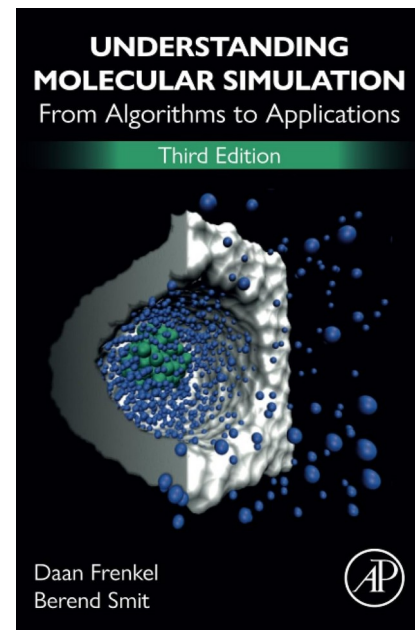
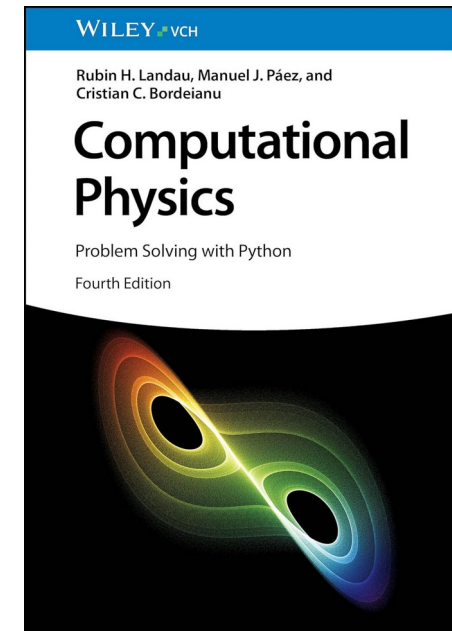
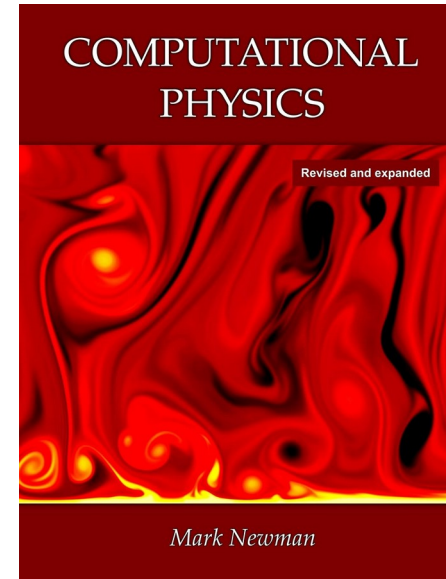
- 20% de Presença e Participação
- 30% de Listas de Exercícios (~06 listas)
- 50% de Projeto Final

Ementa

1. Introdução ao Python
2. Interação de Lennard-Jones
3. Ensemble NVE e integrador de Verlet
4. Ensemble NVT e integrador de Langevin
5. Ensemble NPT
6. Água e Campos de Força
7. Solute e Solvente
8. Integração Termodinâmica

Referências

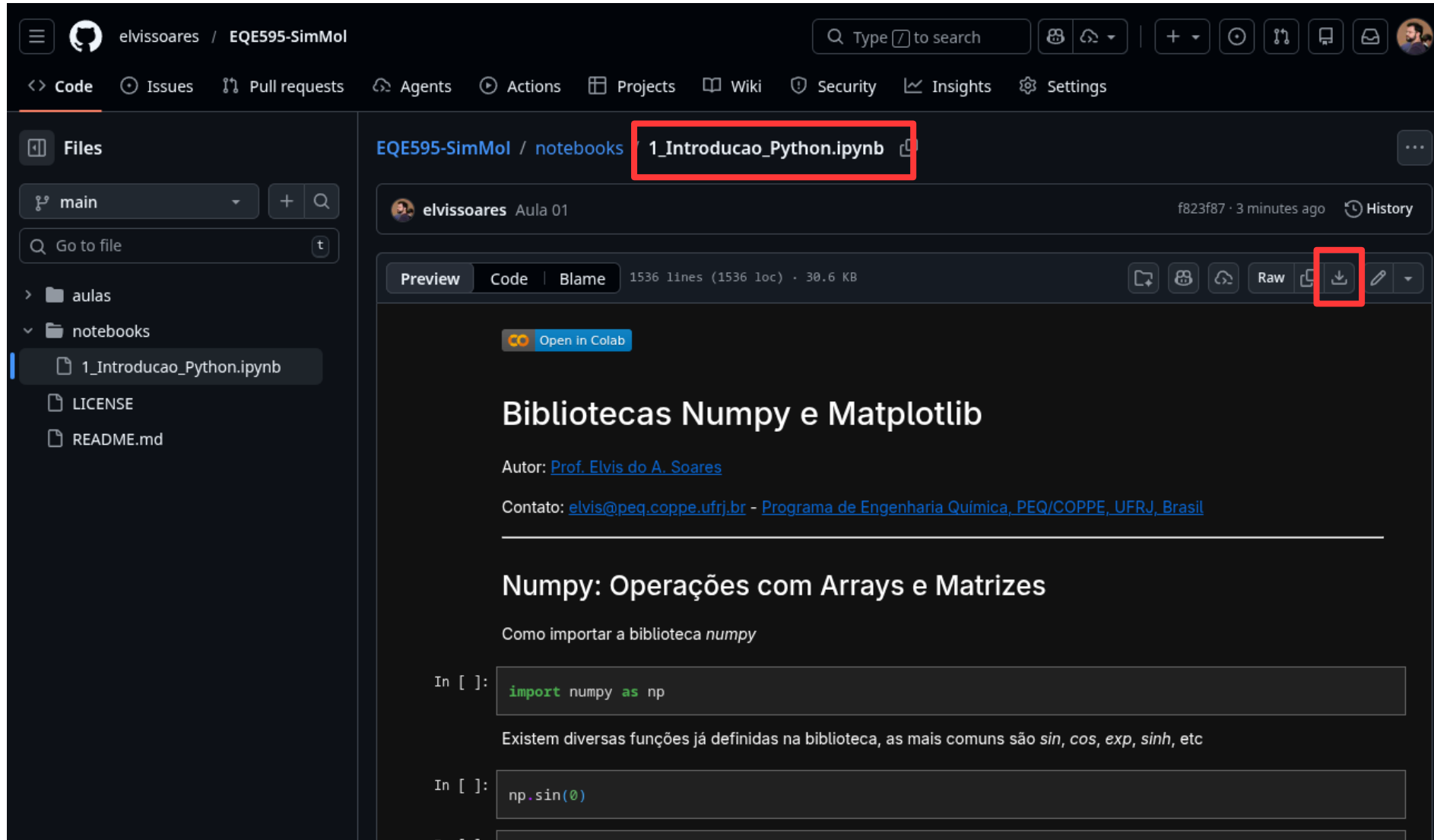
- Newman, M. (2013). **Computational physics**. CreateSpace Independent Publishing Platform.
- Landau, R. H., Páez, M. J., & Bordeianu, C. C. (2024). **Computational physics: Problem solving with Python, 4th Edition**. John Wiley & Sons.
- Frenkel, D., & Smit, B. (2023). **Understanding molecular simulation: from algorithms to applications, 3rd Edition**. Elsevier.
- Allen, M. P., & Tildesley, D. J. (2017). **Computer simulation of liquids, 2nd Edition**. Oxford university press.



Ferramentas Computacionais



Baixando o Notebook



The screenshot shows the GitHub interface for the repository `elvissoares / EQE595-SimMol`. The `Code` tab is selected. In the `Files` sidebar, the `notebooks` folder is expanded, and the file `1_Introducao_Python.ipynb` is highlighted. The main content area shows the notebook's preview, titled `1_Introducao_Python.ipynb`. The notebook content includes a title `Bibliotecas Numpy e Matplotlib`, author information `Autor: Prof. Elvis do A. Soares`, contact information `Contato: elvis@peq.coppe.ufrj.br - Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil`, and a section titled `Numpy: Operações com Arrays e Matrizes`. The notebook content also includes a code cell with `import numpy as np` and a text block explaining that various functions are defined in the library, such as `sin`, `cos`, `exp`, `sinh`, etc. The download icon (a downward arrow) in the top right of the notebook preview area is highlighted with a red box.

elvissoares / EQE595-SimMol

Code Issues Pull requests Agents Actions Projects Wiki Security Insights Settings

Files

main

Go to file

aulas

notebooks

1_Introducao_Python.ipynb

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elvissoares Aula 01 f823f87 · 3 minutes ago History

Preview Code Blame 1536 lines (1536 loc) · 30.6 KB

Open in Colab

Bibliotecas Numpy e Matplotlib

Autor: [Prof. Elvis do A. Soares](#)

Contato: elvis@peq.coppe.ufrj.br - Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil

Numpy: Operações com Arrays e Matrizes

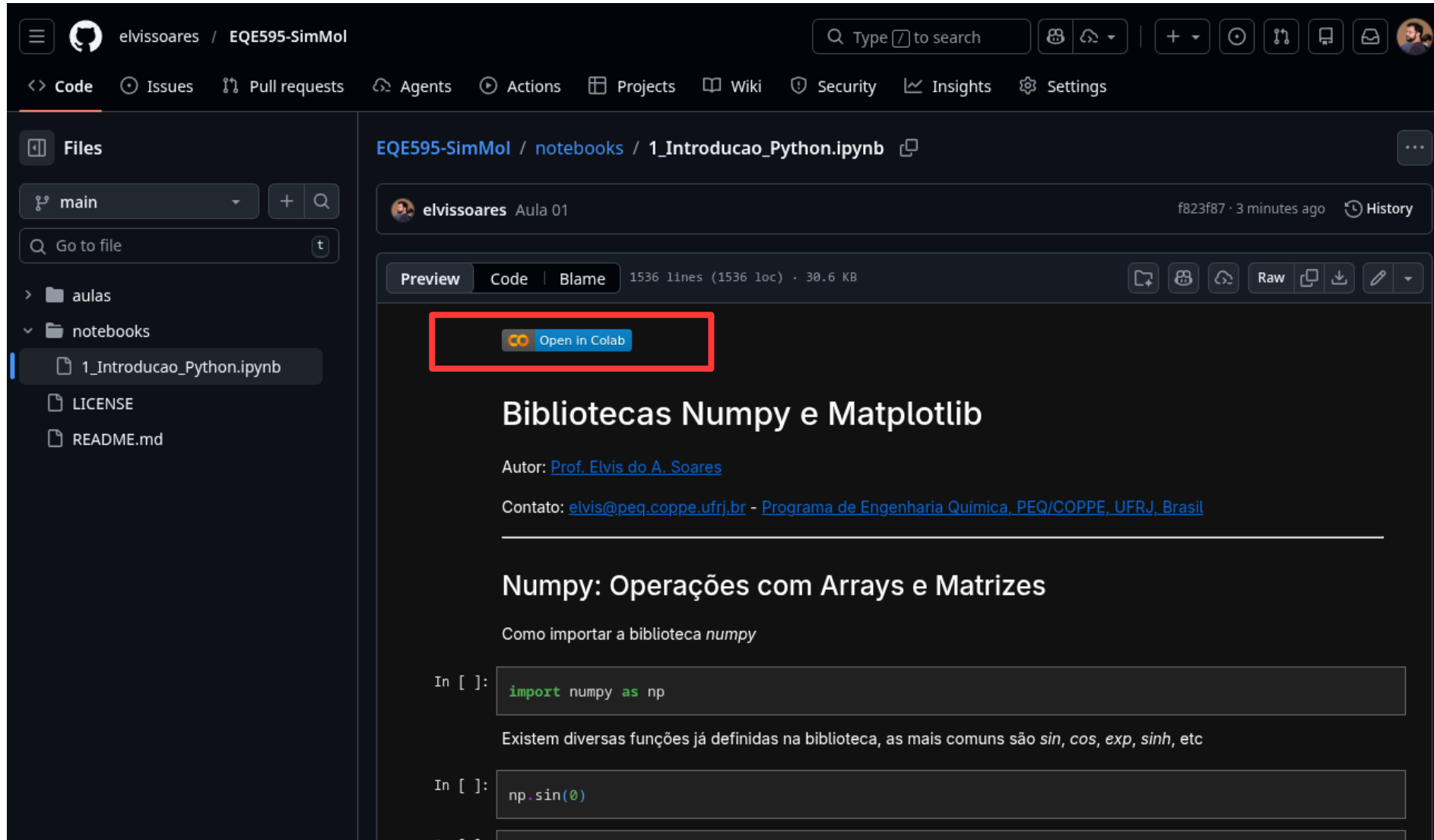
Como importar a biblioteca `numpy`

```
In [ ]: import numpy as np
```

Existem diversas funções já definidas na biblioteca, as mais comuns são `sin`, `cos`, `exp`, `sinh`, etc

```
In [ ]: np.sin(0)
```

Abrindo o Notebook no Colab



The screenshot shows the GitHub interface for the repository `elvissoares / EQE595-SimMol`. The left sidebar displays the file structure with folders `aulas` and `notebooks`, and files `1_Introducao_Python.ipynb`, `LICENSE`, and `README.md`. The main area shows the notebook `1_Introducao_Python.ipynb` in the `Preview` tab. A red box highlights the `Open in Colab` button. The notebook content includes a title `Bibliotecas Numpy e Matplotlib`, author information, and code cells for importing `numpy` and using `np.sin()`.

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Code Issues Pull requests Agents Actions Projects Wiki Security Insights Settings

Files

main

Go to file

aulas

notebooks

1_Introducao_Python.ipynb

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EQE595-SimMol / notebooks / 1_Introducao_Python.ipynb

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Bibliotecas Numpy e Matplotlib

Autor: [Prof. Elvis do A. Soares](#)

Contato: elvis@peq.coppe.ufrj.br - Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil

Numpy: Operações com Arrays e Matrizes

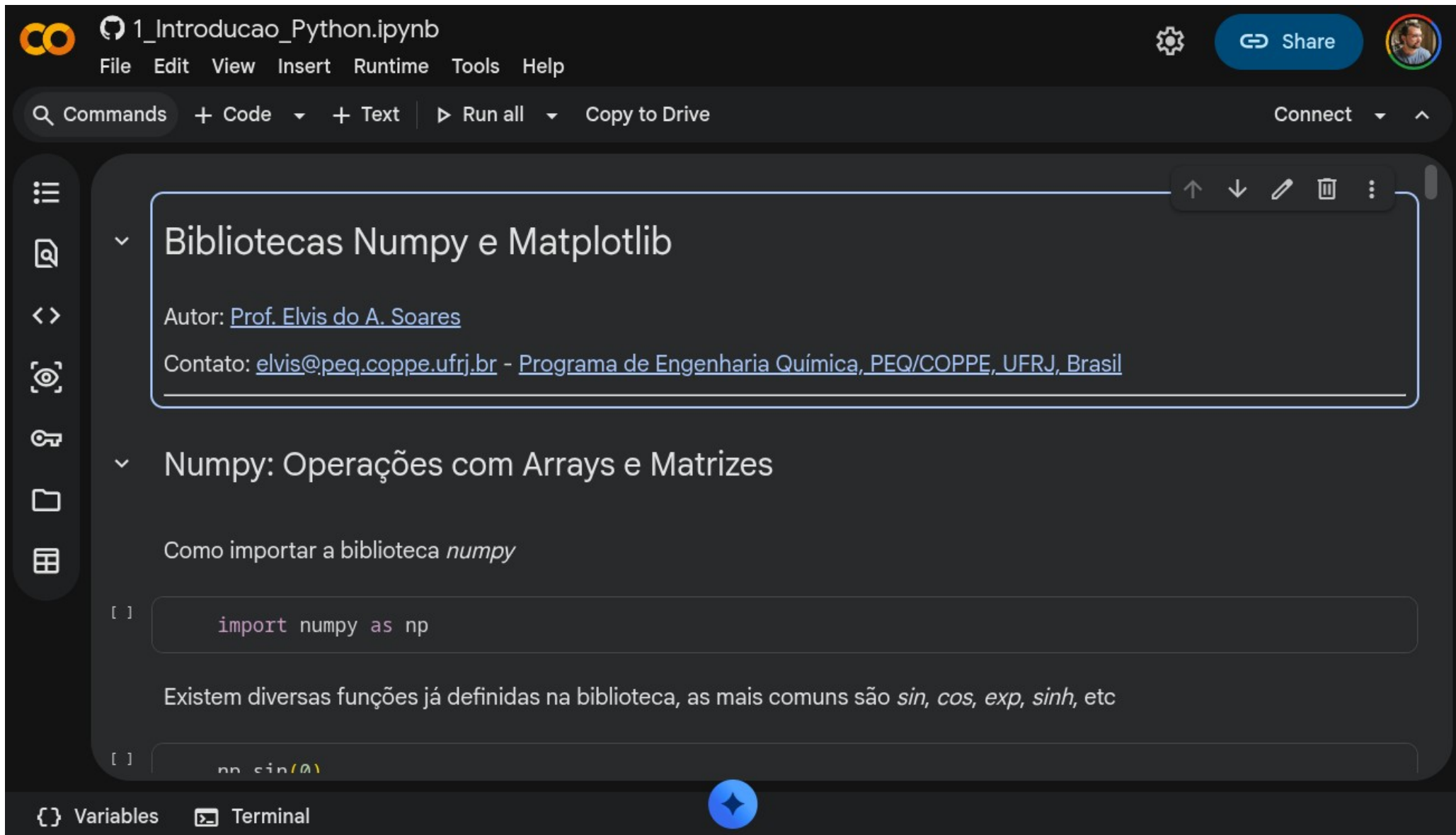
Como importar a biblioteca `numpy`

```
In [ ]: import numpy as np
```

Existem diversas funções já definidas na biblioteca, as mais comuns são `sin`, `cos`, `exp`, `sinh`, etc

```
In [ ]: np.sin(0)
```

Usando Python pelo Colab



The screenshot shows the Google Colab interface for a notebook titled "1_Introducao_Python.ipynb". The top bar includes the Colab logo, a menu (File, Edit, View, Insert, Runtime, Tools, Help), a settings gear, a "Share" button, and a user profile picture. Below the top bar is a toolbar with "Commands", "Code", "Text", "Run all", and "Copy to Drive" buttons. The left sidebar contains icons for file management and a table of contents. The main content area displays two sections:

- Bibliotecas Numpy e Matplotlib**
 - Autor: [Prof. Elvis do A. Soares](#)
 - Contato: elvis@peq.coppe.ufrj.br - [Programa de Engenharia Química, PEQ/COPPE, UFRJ, Brasil](#)
- Numpy: Operações com Arrays e Matrizes**
 - Como importar a biblioteca *numpy*
 - []

```
import numpy as np
```
 - Existem diversas funções já definidas na biblioteca, as mais comuns são *sin*, *cos*, *exp*, *sinh*, etc
 - []

```
np.sin(0)
```

The bottom of the interface shows tabs for "Variables" and "Terminal", and a blue circular button with a star icon.